Step 1 – Measure and Cut the Cable

First, I measured the required length of the Ethernet cable. After measuring, I used the cutter on the crimping tool to cut the cable neatly.

A clean cut makes the next steps much easier.

Step 2 – Strip the Outer Jacket

Next, I used the wire stripper to remove about **2 to 3 cm (1 inch)** of the outer plastic jacket.

While removing the jacket, I made sure not to damage the insulation of the eight small wires inside.

If any wire gets cut or scratched, the cable should be cut again and stripped properly.

Step 3 – Separate the Twisted Pairs

Inside the cable there were four twisted pairs.

- Orange pair
- Green pair
- Blue pair
- Brown pair

I carefully untwisted each pair only as much as needed because keeping the twists close to the connector helps reduce signal interference.

Step 4 – Arrange the Wires

I straightened all eight wires using my fingers and arranged them in the **T568B standard**.

The order was:

White/Orange

Orange

White/Green

Blue

White/Blue

Green

White/Brown

Brown

I checked the color order twice before moving to the next step because even one wire in the wrong position will make the cable fail.

Step 5 – Trim the Wires

After arranging the wires, I held them tightly together and cut all eight wires to the same length.

The cut had to be perfectly straight so that every wire would reach the end of the RJ-45 connector equally. Uneven wire lengths are a common cause of failed crimps.

Step 6 – Insert the Wires into the RJ-45 Connector

I held the RJ-45 connector with the clip facing downward.

Then I slowly pushed all eight wires into the connector while making sure:

- Every wire entered its correct channel.
- Every wire reached the front end of the connector.
- The outer cable jacket also entered slightly inside the connector so it could be secured by the strain relief.

I looked through the transparent connector to confirm the wire colors were still in the correct order.

Step 7 – Crimp the Connector

After confirming everything was correct, I inserted the RJ-45 connector into the crimping tool.

I squeezed the handles firmly until the ratchet released.

This pushed the gold pins into the wires and locked the connector onto the cable.

Step 8 – Test the Cable

Finally, I connected both ends of the cable to a LAN cable tester.

The tester checked all eight pins.

If the LEDs lit up in sequence from **1 to 8**, it meant the cable had been crimped successfully.

If any LED did not light up or appeared in the wrong order, the connector had to be cut off and crimped again.

Common Mistakes to Avoid

1. Wrong Wire Color Order

One of the most common mistakes is placing the wires in the wrong sequence.

Even if only one wire is misplaced, the Ethernet cable may not work or may operate at a lower speed.

How to avoid it:

Always verify the T568B color order before inserting the wires into the connector.

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2. Wires Not Fully Inserted

Sometimes the wires stop before reaching the front of the RJ-45 connector.

When this happens, the gold pins cannot make proper contact with the copper wires.

How to avoid it:

Check through the transparent connector to make sure every wire touches the front before crimping.